4	**(a)**	**(i)**	when two (or more) waves meet (at a point)	M1	
			there is a change in overall intensity / displacement	A1	
		(ii)	constant phase difference (between waves)	B1	[3]
	(b)	**(i)**	$d\sin\theta = n\lambda$	B1	
			$(10^{-3} / 550)\ \sin 90 = n \times 644 \times 10^{-9}$	C1	
			$n = 2.8$	C1	
			so two orders	A1	[4]
			(*power-of-ten error giving 2800 orders, allow 1/3 only for calculation of n*)		
		(ii)	**1.** $d\sin\theta = n\lambda$ (*either here or in* **(i)** *– not both*)		
			<u>θ is greater so</u> λ is greater	B1	[1]
			2. when n is larger, <u>$\Delta\theta$</u> is larger	M1	
			so greater in second order	A1	[2]

5	**(a)**		metal: crystalline / lattice / atoms in regular pattern	B1	